



TENET

TEAM · SMART INDIA HACKATHON 2026



**SMART INDIA
HACKATHON
2026**

SMART INDIA HACKATHON 2026 · GRAND SUBMISSION

AYUSH-SETU

One Platform · One Ecosystem for skill mapping, AI-proctored NSQF assessments, internships & placements across the six Ayush systems

Ayurveda

Yoga & Naturopathy

Unani

Siddha

Sowa-Rigpa

Homoeopathy

PROBLEM STATEMENT ID

SIH26044

PROBLEM STATEMENT TITLE

Portal for Academia–Industry collaboration for Skill Mapping, Internships and Placement

ORGANISATION

Ministry of Ayush

TEAM NAME

Tenet

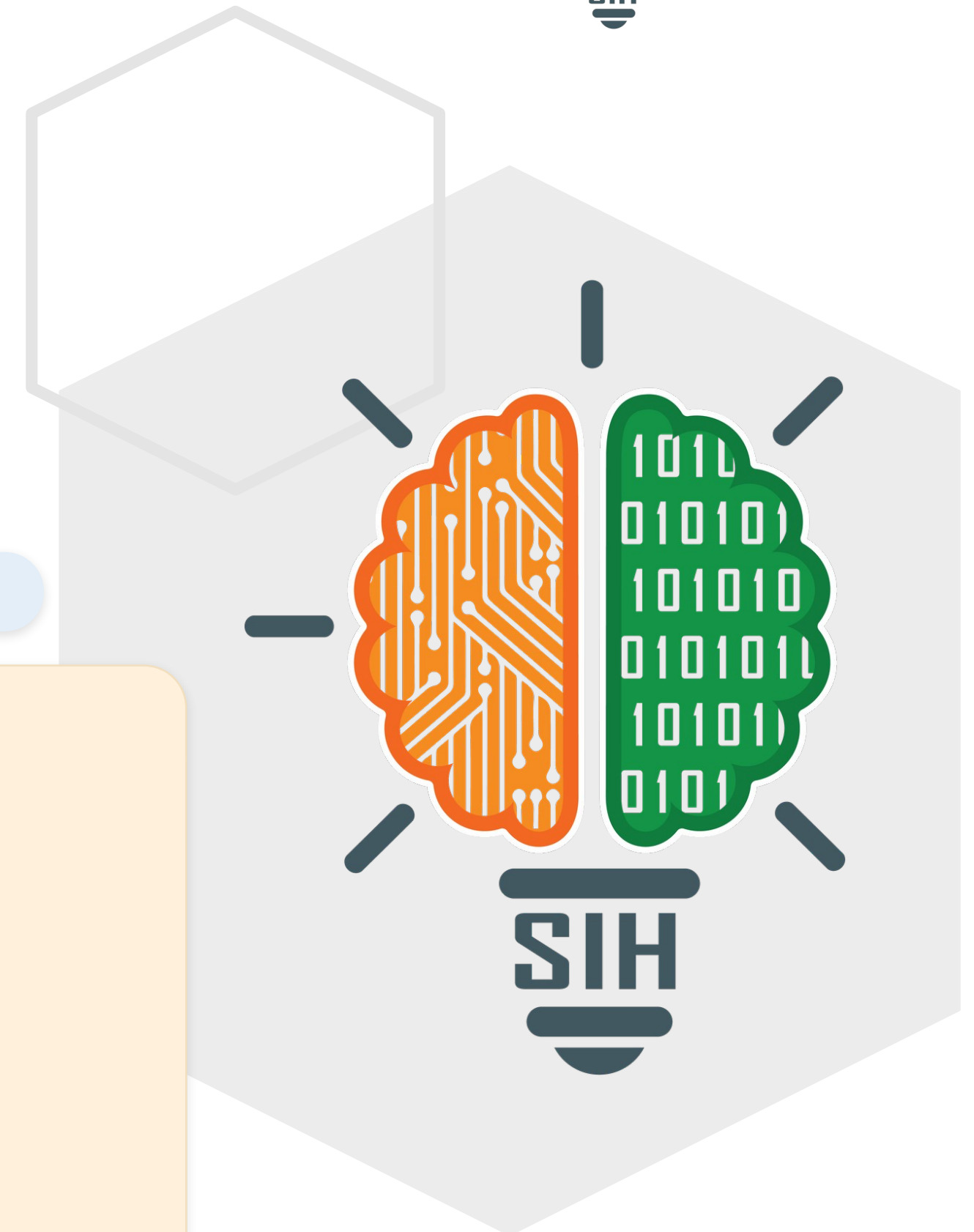
THEME

Smart Automation

PS CATEGORY

Software

TEAM ID



Proposed Solution

A unified web platform (installable PWA) connecting AYUSH students, academicians, institutions and industry across skill mapping, NSQF-verified assessments, internships and placements.

ONE PLATFORM, ONE ECOSYSTEM



Skill Intelligence Layer

NSQF level-wise tests (Levels 1–8) on HSSC Ayush Qualification Packs — Ahar & Poshan Sahayak, Panchakarma Technician, Ayurveda Dietician — build verified skill profiles, bar-chart gap analysis and PDF reports.



Trusted AI-Proctored Assessments

Create tests two ways — upload CSV/JSON/manual or AI-generated level-wise tests — and run them with camera, microphone and entire-screen AI proctoring plus live proctors.



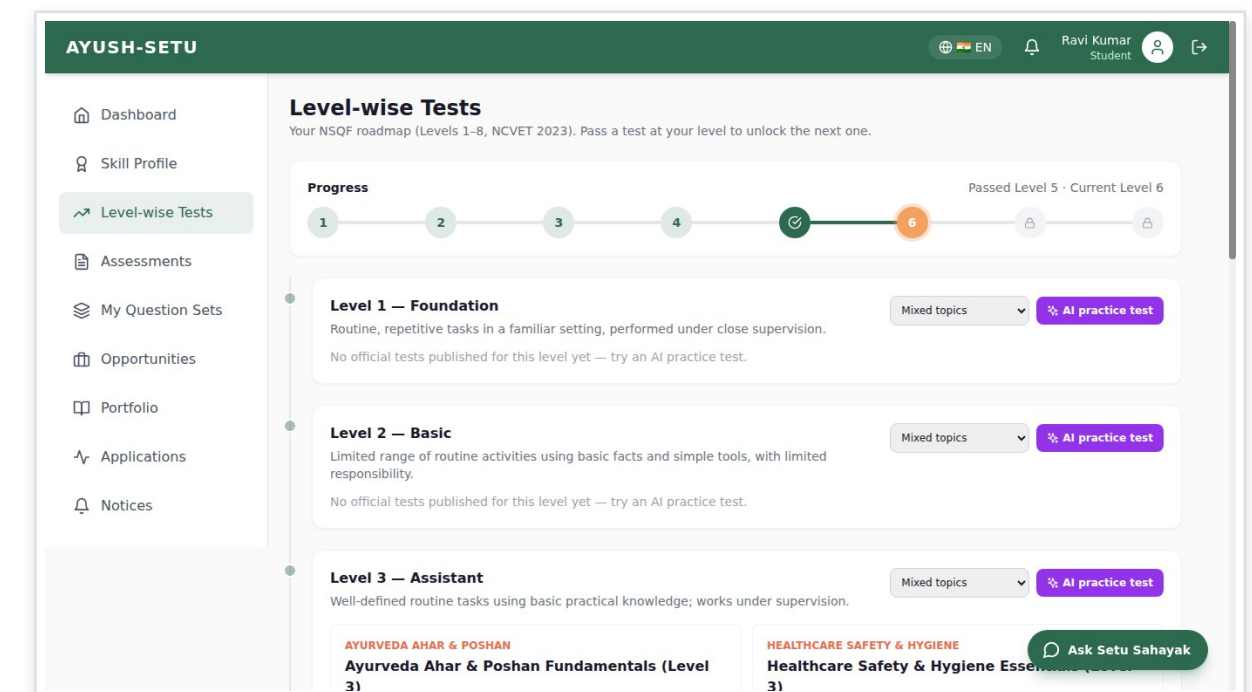
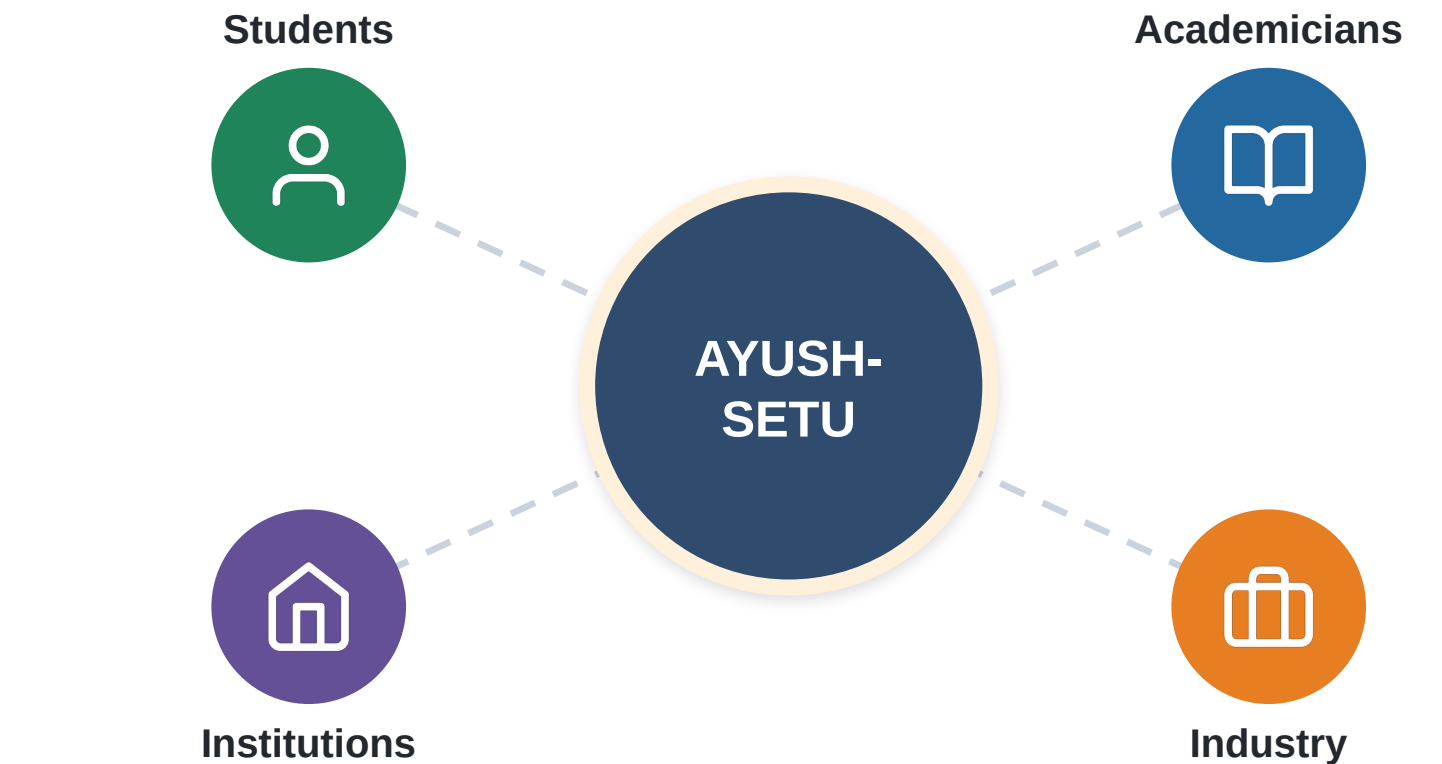
Opportunity & Collaboration Hub

Industry posts internships & jobs matched by AI to verified scores; academicians mentor students; institutions track readiness and placement outcomes.



Portfolio, Notices & Setu Sahayak

Certificates, projects, application tracking, notices and a role-aware chatbot that guides users to features and answers doubts.

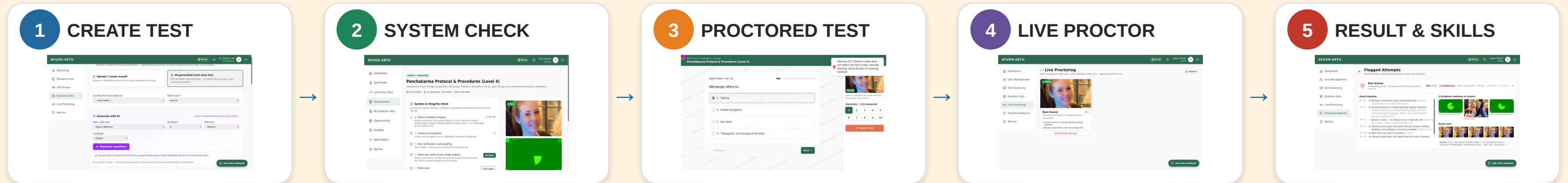


Live prototype: NSQF Level-wise Tests roadmap

Why it's different: an Ayush-focused skill-to-placement portal where every skill score comes from an AI-proctored, NSQF level-wise test — so industry can trust the shortlist.

Technical Approach

MODULAR, SCALABLE & EXPLAINABLE ARCHITECTURE



AI ANTI-CHEATING ENGINE

Keyboard + Screen Sync (remote access)

Entire-screen share with a rotating colour sync marker; keyboard/mouse input is checked against the verified face on camera; AnyDesk/RDP/VM display & cursor-teleport detection; extra monitors blocked.

Voice Monitoring

On-device speech detection against calibrated room noise; a voice while the candidate's lips don't move = another person helping; mic-off alert.

Camera: Extra Person & Phone

Multi-face + person detection, mobile-phone detection (photos for AI help), look-down tracking, face recognition and anti-spoofing.

No Tab Switch / Background Apps

Forced full screen until submit; tab, app & focus tracking; questions hidden when another app takes focus; screen snapshot as evidence.

Other Cheating Methods Closed

Copy/paste, print, shortcuts, DevTools, AI extensions & bots blocked; per-student question/option shuffle; watermark; chatbot off; one live attempt.

Enforcement & Evidence

5 serious violations → auto-submit + flag; live proctor can terminate; server timer & autosave; timeline with webcam/screen snapshots.

TESTS, LEARNING & ASSISTANT

Two ways to create tests

Upload CSV/JSON/manual OR AI-generate (Gemini/OpenAI + offline bank); review, then publish

NSQF Level-wise Tests

Levels 1–8 roadmap, unlock next level on pass, AI practice test per level

Setu Sahayak Chatbot

Takes students, industry & admin to tests, notices, jobs; answers doubts; off during tests

Working tests & bar-graph results

Camera on at start, full-screen test, instant score, skill bar graph vs benchmark, notices

IMPLEMENTATION STACK

Frontend

React 18 • Vite • Tailwind (PWA)

Backend

Node.js • Express • Prisma • FastAPI

AI Layer

TensorFlow.js (Human) • Web Audio • Gemini/OpenAI

Database

PostgreSQL + pgvector

Security

JWT + RBAC • server timer • HTTPS

Methodology: Agile sprints → Working prototype → Pilot with institutions → Feedback loop → Scale through APIs & integrations

FEASIBILITY AND VIABILITY

FEASIBLE TODAY. DESIGNED TO SCALE TOMORROW.

WHY IT IS FEASIBLE

✓ Working prototype already built

✓ Open-source stack · free-tier cloud

✓ Only a laptop with webcam & mic

✓ AI keys optional · offline fallback

LOOPHOLES WE FOUND → HOW AYUSH-SETU CLOSES THEM

● BUILT

● ROADMAP

1 Remote helpers & second screens BUILT

AnyDesk-style control or an extra monitor lets someone else see and answer.

HOW WE SOLVE IT

Entire-screen share with a colour screen-sync marker, display count, remote-display & cursor checks; clicks with no face on camera are flagged.

2 Copied & photographed questions BUILT

Questions get copied, screenshotted or photographed with a phone.

HOW WE SOLVE IT

Per-student question & option shuffle, copy/print/shortcut blocking, traceable watermark and mobile-phone detection.

3 Help from someone in the room BUILT

A friend sits nearby or whispers answers off-camera.

HOW WE SOLVE IT

Multi-face & extra-person detection and voice detection matched with the candidate's lip movement.

4 Impersonation BUILT

A different person sits the test or holds up a photo.

HOW WE SOLVE IT

Face recognition against the face verified at start, plus anti-spoofing and liveness models.

5 Too little test content BUILT

Faculty and industry lack time to write level-wise question banks.

HOW WE SOLVE IT

Upload CSV/JSON/manual or AI-generate a level-wise test in minutes; offline reviewed AYUSH question bank.

6 Low-end laptops & weak internet BUILT

Heavy AI or a dropped connection can break an online test.

HOW WE SOLVE IT

Adaptive AI frame rate, heartbeat autosave, server-side timer and resume with the attempt recorded.

7 MCQs can't prove clinical skill ROADMAP

A web quiz cannot show a student can treat a patient.

HOW WE SOLVE IT

Proctored theory + mentor feedback today; clinic-observed Guru-Shishya endorsements next.

8 Fake certificates ROADMAP

Students can upload unverified certificates.

HOW WE SOLVE IT

Skill scores come only from proctored tests; DigiLocker / ABC credential verification in the next phase.

PHASED EXECUTION ROADMAP

MVP ✓

Proctored level-wise tests, profiles, AI test creation

PILOT

Institution onboarding, matching + applications

EXPAND

DigiLocker/ABC verification, mentor endorsements

SCALE

Analytics, lockdown desktop client, more languages

IMPACT AND BENEFITS

FROM SKILL VISIBILITY TO EMPLOYABILITY OUTCOMES



STUDENTS

- Verified NSQF level evidence
- AI practice tests for every unlocked level
- Clear skill gaps & PDF report
- Matched internships & jobs



INDUSTRY

- Shortlists from proctored scores
- Publish own level-wise tests
- Live proctoring with evidence
- Faster, fairer hiring



ACADEMICIANS

- AI-drafted tests in minutes
- Mentoring & feedback
- Research hub & FDP access
- Curriculum insight from gaps



INSTITUTIONS

- Fair online exams at scale
- Readiness dashboards
- Flagged-attempt reports
- Placement outcome tracking

IMPACT FLYWHEEL

Map

Understand capability



Assess

Verify with proctored tests



Learn

Close skill gaps



Match

Connect to opportunities



Measure

Improve outcomes

8

NSQF levels on one roadmap

44

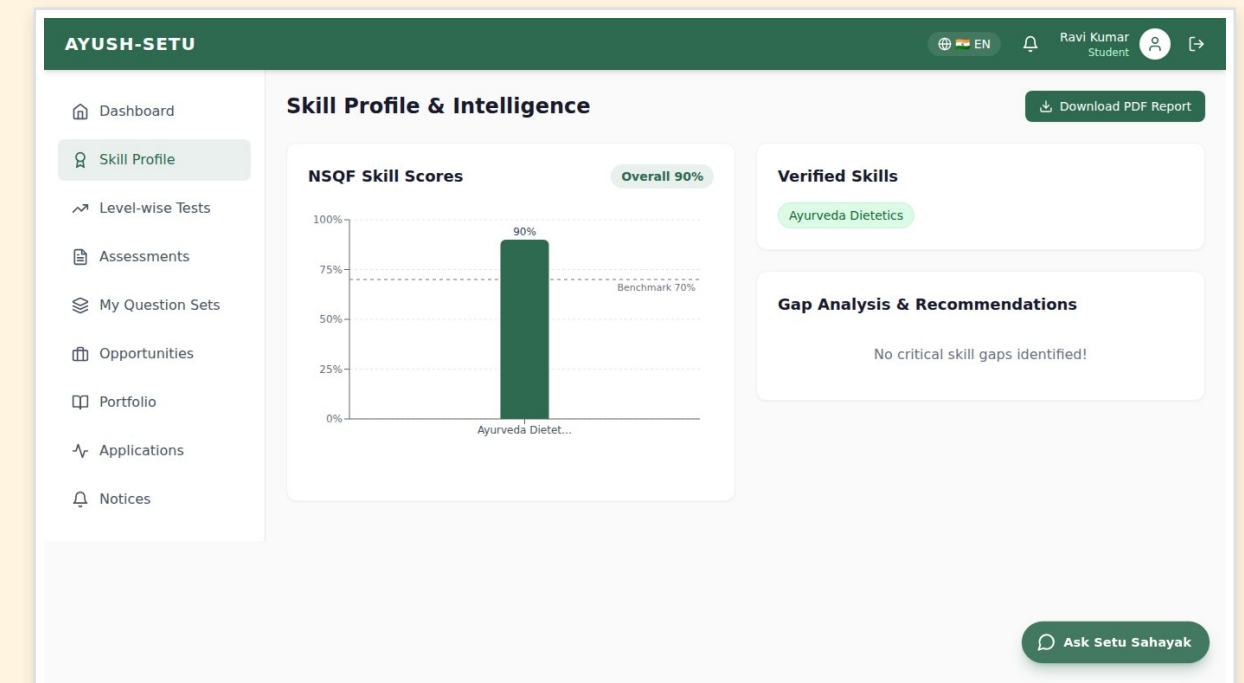
integrity signals per proctored test

2

ways to create tests: upload or AI

0

video streams uploaded (on-device AI)



Skill Profile: verified scores vs 70% benchmark with gap analysis

Research & References

RESEARCH BASIS



SIH 2026 – Problem Statement SIH26044

Ministry of Ayush: Portal for Academia–Industry collaboration for Skill Mapping, Internships and Placement.

sih.gov.in



Ministry of Ayush

The six Ayush systems and national digital initiatives relevant to Ayush education and employment.

ayush.gov.in



NCVET – NSQF 2023

Revised NSQF notified on 6 June 2023: eight levels with descriptor domains that drive our level-wise tests.

ncvet.gov.in



HSSC & National Qualifications Register

Ayush qualification packs, e.g. Panchakarma Technician (HSS/Q3601, NSQF Level 4), used as the skill taxonomy.

healthcare-ssc.in · nqr.gov.in



DigiLocker

Verifiable digital credentials — portfolio KYC today, certificate verification in the next phase.

digilocker.gov.in



On-device AI & Web APIs

Human (TensorFlow.js) face, anti-spoofing and object models; Screen Capture & Window Management APIs.

github.com/vladmandic/human